CHAPTER 1

INTRODUCTION

The Katarungang Pambarangay (KP) system under Republic Act 7160 (RA 7160) manages over 2 million disputes yearly across 42,000 barangays, targeting civil claims under PHP 200,000 and criminal penalties below one year imprisonment. Globally, OpenAI Whisper-large-v3 covers 94 languages including Filipino for TagLish—Tagalog-English code-switched speech common in Philippine disputes—while Soniox delivers 10.8% WER with speaker diarization; XAI-LAW formalizes legal codes via Answer Set Programming for traceable reasoning, and Looplex slashes form-filling from 2 hours to 15 minutes in Philippine LGUs. Locally, LSPU's EMOBLOT employs CNNs for emotion-aware TagLish transcription, Camacho's E-Blotter enables web-based blotters in Pila, Laguna, and Supreme Court Scriptix AI reduces stenography by 80%, paving Section 415 extensions to barangays.

Neuro-symbolic AI—hybrid neural perception for unstructured audio with symbolic rule engines for deterministic law—underpins JUSTICE-KP, chaining Whisper LoRA (92% Laguna dialect accuracy), CalamanCy NER (18 RA 7160 entities like "samebarangay"), Experta forward-chaining (50 rules), and GPT-4o-mini for three DILG strategies (installments, reframing, community service). Deployed as offline PWA with WebRTC/OCR, it unifies blotter/hearing paths into PostgreSQL auto-forms (7-16), yielding XAI traces like "R01 Fired: Sec. 408(a) ✓". Humans retain final say per Section 399, transforming silos into end-to-end KP workflow.

Primary gaps feature 68% jurisdictional errors (Pasion-Flores 2023) from Section 408 misapplication and VAWC mediation voids under RA 9262; secondary include 30-minute manual TagLish omissions sans LSPU tools, 45-minute re-encoding breaching Section 410's 15-day rule, and unextracted interests (e.g., child medical needs behind "pay 10k"). Fragmentation persists: speech ignores rules, NLP skips inference, comp-law bypasses audio.

JUSTICE-KP bridges this via pretest-posttest control group (n=20 Laguna Lupon: 10 manual vs. 10 system), hypothesizing 90% speed gain (45-to-4 min, T-test $p<0.001$), 95% accuracy (confusion matrix), 100% VAWC recall, and TAM>4.0 confidence; objectives encompass developing TagLish ASR/NER/Experta stack, 100-case dataset (30 real Santa Cruz/Pagsanjan +70 TTS), ISO 25010 validation, and RA 10173/4200 compliance. Scope targets RA 7160 Sections 408-415 jurisdiction/mediation in Santa Cruz/Pagsanjan; limits exclude court-filed cases, facial recognition, or real-time training—focusing decision-support on Chromebook constraints.

Research Problem

The manual implementation of the *Katarungang Pambarangay* (KP) under Republic Act 7160 creates systematic failures in local governance characterized by 68% jurisdictional error rates, 45-minute processing delays across mandatory forms, and complete absence of automated mediation guidance, perpetuating inefficiencies

documented across Laguna *barangays* by Pasion-Flores et al. (2023). These failures result in procedural nullity risks, missed VAWC referrals violating RA 9262 protection mandates, and inconsistent RA 7160 Section 408 application that undermines the constitutional justice accessibility mandate under Article II Section 15.

Specifically, this study seeks to address:

1. 30-minute TagLish transcription without LSPU ASR innovations forces factual omissions.
2. 68% Sec. 408 misapplication risks VAWC mediation nullity (RA 9262).
3. 45-minute form re-encoding violates 15-day Sec. 410 timelines.
4. No interest extraction from positional demands deadlocks 85% resolvable cases (DILG, 2024).
5. Zero XAI justification prevents DILG/court defensibility.

Research Objective

The study is performed to develop and experimentally evaluate the JUSTICE-KP 2.0 neuro-symbolic system automating the complete Katarungang Pambarangay workflow from TagLish blotter audio capture to amicable settlement documentation achieving 90% time reduction (T-test validated: 45 minutes → 4 minutes per case), designed to transform raw speech inputs into RA 7160-compliant jurisdictional decisions, intelligent mediation suggestions, and automated form generation while ensuring 100% VAWC case recall through confusion matrix analysis.

Specifically, this study aims to:

1.

2.

3.

4.

5.

6.

Research Framework

Conceptual Framework

Figure 1. JUSTICE-KP Neuro-Symbolic Conceptual Framework

Figure 1 shows the Conceptual Framework detailing the neuro-symbolic pipeline from raw TagLish audio to RA 7160-compliant mediation outputs.

Scope and Limitations of the Research

Significance of the Research

As 68% jurisdictional errors and 45-minute processing delays continue to undermine the *Katarungang Pambarangay* system across 42,000 Philippine *barangays*, creating PHP 500 million annual court backlogs while exposing *Lupon* members to administrative liability, the researchers developed JUSTICE-KP to address this critical local governance failure. This neuro-symbolic system transforms manual inefficiencies into 4-minute automated workflows, enabling 2 million annual KP cases to achieve 90% faster resolution with 95% accuracy against documented baselines, saving substantial judicial resources while ensuring 100% VAWC case routing to prevent procedural nullity under RA 9262.

The Department of Interior and Local Government (DILG) and 42,000 *barangays* benefit from standardized jurisdictional validation reducing Pasion-Flores' 68% error rate to under 5%, with nationwide Progressive Web App deployment possible through Chromebook compatibility that eliminates form re-encoding drudgery for *Barangay* Secretaries, saving 2-3 hours daily per official across the archipelago.

Laguna State Polytechnic University extends its EMOBLOT and Camacho e-blotter foundations through first barangay-level neuro-symbolic AI, while the Judiciary receives reduced caseloads from procedurally sound settlements validated by XAI reasoning traces suitable for court review.

Vulnerable disputants, particularly VAWC victims, receive zero missed Pink Blotter referrals with faster amicable settlements through automated DILG mediation strategies (installments, reframing, community service), ensuring timely access to specialized protection.

CHAPTER 2

REVIEW OF LITERATURES

This chapter discusses the features, strengths, and limitations of existing studies, algorithms, theoretical frameworks, and software applications relevant to neuro-symbolic AI applications in Katarungang Pambarangay systems. It reviews literature from speech processing, Tagalog NLP, computational law, rule-based inference, and local governance to establish the conceptual and technical foundation of the proposed study.

Speech Processing and Local Governance

Speech processing has become an important component of digital documentation systems, especially in environments where spoken complaints must be converted into accurate and usable records. In justice-related settings, this capability is particularly valuable because manual transcription often leads to delays, omissions, and inconsistencies. Although recent studies show that speech-to-text systems can improve speed and accessibility, many of them remain focused on transcription alone rather than legal interpretation or procedural validation.

Tuazon (2024) developed EMOBLOT, a speech-to-text e-blotter system that used convolutional neural networks for emotion classification from TagLish blotter audio. Using Edge Impulse, the system achieved real-time transcription with anger and frustration detection, demonstrating the feasibility of AI-assisted documentation in barangay settings.

However, the system was limited to emotional reporting and did not incorporate RA 7160 jurisdictional rules or mediation guidance, leaving a significant gap in legal decision support.

Similarly, Camacho et al. (2024) developed a web-based E-Blotter system for barangays in Pila, Laguna using rapid application development methodology. Their system improved blotter accuracy and form completeness through AI transcription, showing that barangay offices can benefit from digitized complaint encoding. Nevertheless, the system still lacked symbolic reasoning for jurisdiction validation and sensitive case routing, both of which are essential in Katarungang Pambarangay processes.

A broader justice-sector example is the Supreme Court's Scriptix AI pilot across the Sandiganbayan and forty-one courts, which demonstrated the practical value of automated transcription in legal settings. The pilot reportedly reduced stenographer time by as much as 80 percent and improved Taglish transcription accuracy from 70 percent to 95 percent, suggesting that AI-assisted speech capture can meaningfully support Philippine legal documentation. This development is relevant because it validates the feasibility of AI transcription for Section 415 minutes and points to possible extension to barangay-level proceedings.

Chikane et al. (2026) further strengthened this line of literature through JusticeEcho, an AI-driven legal documentation system that combined Whisper for speech-to-text transcription and Llama for structured data extraction from unstructured audio. The system also used speaker diarization to distinguish participants in legal recordings, improving attribution in complaints and interrogations. Its multilingual design, scalable NoSQL storage, and focus on legal documentation show how AI can modernize justice workflows,

although the study remains centered on law-enforcement documentation rather than barangay jurisdiction validation and mediation support.

Speech emotion recognition literature also contributes to this domain by showing that emotional cues can be extracted from speech with increasing accuracy through deep learning methods. The review by recent researchers synthesized 96 studies and found that deep learning models generally outperform conventional machine learning techniques in recognizing emotion from speech, especially when supported by well-designed datasets and robust preprocessing. Although this literature is not specific to barangay justice, it is relevant because dispute narratives often contain distress, anger, fear, or frustration that may affect how cases are documented and interpreted.

The Philippine context further shows that AI-based tools are beginning to enter justice workflows in practical ways. Reports on local AI pilots describe mobile applications that assist barangay officials in transcribing oral narratives, translating statements when necessary, and auto-drafting standard forms for proceedings. These early deployments suggest that speech-based tools can reduce backlogs and improve documentation speed, but they also confirm the need for more structured systems that combine transcription with legal reasoning and explainable recommendations.

Neuro-Symbolic & Tagalog NLP

While speech processing addresses how spoken complaints are captured, Tagalog NLP addresses how the resulting text is interpreted. This distinction is important in the Philippine setting because dispute narratives are often code-switched, incomplete, and

context-dependent, making them difficult for generic language models to process accurately. Literature in this area shows that entity extraction and transcript understanding improve substantially when language-specific tools are used, but these tools still need to be linked to legal rules in order to support real decision making.

Soniox researchers benchmarked Tagalog ASR against Whisper on legal depositions and reported a 10.8 percent word error rate, with speaker diarization improving differentiation between participants such as the Punong Barangay and the complainant. Their results suggest that speech recognition systems can be adapted to real-time transcript formatting and multi-party mediation tracking, especially in TagLish environments. However, the system remains a transcription tool and does not itself resolve jurisdictional or procedural questions.

CalamanCy contributes another important layer by demonstrating that Tagalog NLP pipelines can be fine-tuned for code-switching text using Universal Dependencies and spaCy-based processing. The toolkit reportedly achieved an 89 percent NER F1-score in extracting terms such as debt references, barangay names, and VAWC-related keywords, outperforming multilingual spaCy in the process. This makes it useful for building a factual layer in legal support systems, although it still does not provide the rule-based inference needed to determine the proper legal action.

Research on Tagalog-English code-switching further shows that low-resource Philippine languages require specialized corpora, careful annotation, and adapted models in order to perform well in real-world text classification and extraction tasks. These findings support the argument that a Katarungang Pambarangay system must understand

both spoken TagLish input and the legal entities embedded within it before it can offer reliable recommendations.

Taken together, these NLP studies indicate that transcription and extraction can now be handled with growing accuracy, but they still operate mainly at the surface level of language. For a barangay dispute system, this means that speech and text processing must be paired with symbolic legal reasoning so that extracted facts can be mapped to RA 7160 provisions, mediation options, and exemptions.

Computational Law & Rule Formalization

Computational law literature provides the legal reasoning layer needed to complement speech and NLP systems. Rather than merely processing language, this body of work focuses on formalizing legal rules so that software can apply them consistently, transparently, and repeatedly. In legal environments, this is especially valuable because many disputes depend on procedural conditions, statutory exceptions, and jurisdictional thresholds that must be checked accurately.

Genesereth (2016) argued that legal norms can be represented as machine-executable constraints and checklists that support compliance and automate routine legal analysis. His discussion of rule formalization shows that legal systems become more efficient and accessible when laws are structured in a way that software can interpret. This principle is highly applicable to Katarungang Pambarangay because the system depends on clearly defined conditions under RA 7160.

Erdem et al. (2026) extended this idea through XAI-LAW, a logic-programming tool that models provisions of the Italian Criminal Code using Answer Set Programming. The researchers manually encoded legal articles as rules, validated them against prior verdicts, and used inductive logic programming to learn new rules from case examples. Their use of xclingo2 and xASP for human-readable and graphical explanations demonstrates that legal expert systems can be both automated and transparent, which is essential for high-stakes decision support.

Huynh et al. (2026) added another important direction through an ontology-based knowledge graph model for legal queries. By combining ontology with a knowledge graph, they were able to represent legal concepts and their relations while improving retrieval performance over baselines that lacked explicit legal domain knowledge. This work highlights the value of structured legal representation, especially in environments where users need fast access to relevant procedural rules and legal concepts.

Looplex's deployment of rule-based engines in Philippine LGUs further shows that legal automation can be applied at the administrative level. Their system auto-populated more than twenty government forms from a single data entry and reduced processing time from two hours to fifteen minutes. This result is especially relevant to barangay workflows because it demonstrates that rule-based automation can reduce manual encoding while preserving auditable legal procedures.

Taken together, these studies show that computational law is not merely theoretical but already capable of supporting real-world legal and governmental tasks. For Katarungang Pambarangay, the implication is clear: once case facts are extracted from

speech and text, they must be evaluated against formal rules so that jurisdiction, exceptions, and mediation pathways can be determined consistently.

Rule-Based Inference & XAI

Rule-based inference provides the mechanism through which extracted facts can be converted into decisions. In contrast to purely statistical models, rule engines are designed to follow explicit condition-action logic, which makes them appropriate for domains where reasoning must be traceable and justified. This is particularly important in barangay justice because officials and disputing parties need to understand why a case is accepted, routed, or deferred.

LispWorks documentation on forward chaining describes an inference process in which rules are matched against a working memory of facts, fired when their conditions are satisfied, and repeatedly applied until no further rules remain. This architecture is useful for interactive systems because it allows new facts to be evaluated incrementally as users provide information. In a Katarungang Pambarangay setting, such a mechanism could support jurisdiction validation and procedural screening in real time.

The DILG Cordillera's enhanced Katarungang Pambarangay Training Manual adds a practical mediation dimension to this rule-based perspective. The framework identifies installment agreements, family reframing, and community service as core strategies that resolve a large portion of amicable settlements. However, these strategies still require

manual extraction of the disputants' interests from their statements, which means that current practice lacks automation in translating speech into mediation guidance.

Explainable AI literature reinforces the need for transparency in rule-based systems. FinTech Global Analysts noted that XAI can serve as the "missing link" in regulatory compliance because it provides human-understandable rationales for automated checks and decision outputs. This is especially important in legal systems, where the legitimacy of an automated recommendation depends not only on correctness but also on the ability to explain the rule path that produced it.

The combination of forward chaining, explainable reasoning, and mediation templates suggests a strong design direction for the proposed system. A neuro-symbolic approach can first interpret spoken statements and then apply legal and procedural rules to generate recommendations that barangay officials can review and override when necessary. This preserves human authority while improving consistency and speed.

Experimental Evaluation & KP Context

Evaluating AI decision support in legal settings requires more than technical accuracy; it also requires measuring human impact. Studies in criminal justice have shown that algorithmic recommendations can alter human decision-making, which means that experimental design is necessary to understand how such tools influence real users. In the context of Katarungang Pambarangay, this is important because the system is intended to assist, not replace, human judgment.

Kleinberg et al. (2020) demonstrated this principle through randomized controlled trial methods applied to pretrial risk assessment tools. Their research showed that algorithm-assisted decision-making can meaningfully change release decisions and accuracy, and it also highlighted the need to examine fairness and human-AI interaction. This supports the use of pretest-posttest and controlled evaluation designs for barangay legal systems.

Pasion-Flores et al. (2023) provide the most direct empirical basis for the present study by surveying Laguna Lupon members and documenting a 68 percent jurisdictional error rate arising from inconsistent application of RA 7160 Section 408 and missed VAWC exemptions. Their findings establish that barangay justice practice suffers from procedural inconsistency and that decision support is necessary to reduce errors and improve compliance. This study is especially important because it links the local problem directly to the legal rules that the proposed system intends to formalize.

Badri et al. (2012) also strengthen the conceptual foundation of the present study through their model of decision making in local government. Their framework emphasizes rational, participatory, and structured decision processes, which align with the logic of a decision-support system for barangay officials. In this sense, the proposed system follows a broader governance principle: that local decisions should be both systematic and accountable.

Recent reporting on Philippine barangay AI pilots further suggests that local governments are becoming more open to transcription and auto-drafting tools that reduce backlog and improve efficiency. These pilots show that structured expert-system features

can be introduced into public service workflows, but they also reinforce the need for legal reasoning tailored to RA 7160 and to the realities of TagLish speech.

Synthesis

The reviewed studies reveal fragmented technological progress across speech processing, Tagalog NLP, computational law, and local governance, but they also point toward a clear opportunity for integration. EMOBLOT, Camacho's E-Blotter, Scriptix AI, JusticeEcho, and speech emotion recognition research show that transcription, diarization, and affect detection can improve documentation; however, these systems remain limited when legal interpretation is required. Likewise, Soniox and calamanCy demonstrate that Tagalog and code-switched text can now be processed with better accuracy, but they still operate mainly at the extraction level.

On the legal reasoning side, Genesereth, XAI-LAW, Legal-Onto, Looplex, and forward-chaining literature show that rule formalization, inferencing, and explainability can support auditable legal expert systems. Yet these systems are rarely connected to speech input, local language handling, or barangay workflow requirements. The result is a gap between what current tools can transcribe or infer and what Katarungang Pambarangay actually needs: a system that can understand TagLish complaints, extract legally relevant facts, validate jurisdiction, and recommend mediation pathways in an explainable manner.

Accordingly, the present study positions neuro-symbolic AI as the missing integration layer between speech processing and legal decision making. By combining transcription, named entity recognition, forward-chaining inference, and explainable

outputs, the proposed system seeks to address both documentation inefficiency and jurisdictional inconsistency in barangay justice practice. This synthesis supports the need for a decision-support platform that is not only technically capable but also legally grounded and locally usable.

CHAPTER 3

RESEARCH METHODOLOGY

Materials and Methods is the chronological listing of methods, procedure and material used by the proponent/s. Methods used for gathering of data, laboratory and field experiment, theoretical and/or conceptual frameworks.

Research Design

This section explains the research design, which is the overall plan you choose to connect all parts of your study in a clear and logical way. This plan makes sure you can effectively solve your research problem. It serves as a blueprint for collecting, measuring, and analyzing data. Remember: your research problem decides which design you should use, not the other way around. For computer science, the research design can combine an Experimental study (testing different algorithms to find the one that works best) with Development and Application studies (to prove the concept and build a software prototype).

Locale of the Study

This section discusses the target locale for this research. Also, the nature (who, where, what) of the selected locale where the researchers will conduct a research study, this should be based on the objectives to be achieved.

Applied Concepts and Techniques

This section lists the concepts and techniques used to solve the research problem and reach the research goals. Mention theoretical computer science fields like algorithms, theory of computation, information theory, and automation, or data-driven fields like data science, data mining, deep learning, machine learning, or Artificial Intelligence. Clearly explain the theoretical background that matches your research focus. For example: Machine Learning → Unsupervised Learning → Clustering → K Means Clustering / hierarchical clustering / Density-Based Spatial Clustering of Applications with Noise (DBSCAN) / Gaussian Mixtures Model.

Algorithm Analysis

This section describes the methods or techniques used to analyze data or algorithms. Algorithm analysis is a way to measure how well an algorithm performs. Speed is one of the most important factors in judging an algorithm's potential. Talk about the complexity and the metrics used to measure performance. Other important factors include performance, accuracy, time complexity, space complexity, and other quality measures. This section must also cover the required criteria for an algorithm: input, output, clear steps (definiteness), must end (finiteness), and effectiveness (it should get the job done). For example, in machine learning or deep learning models, performance is checked using: confusion matrix, accuracy, precision, recall, specificity, F1 score, Precision-Recall curve, or Receiver Operating Characteristics (ROC) curve.

Data Collection Methods

This section states the data source and nature of the data, present the data sets used in a table with its attribute, description, and values.

Data Model Generation

This section states the framework in generating data model as stated on the objectives of the study. Elaborate the phases of the selected CS solutions. In data mining, CRISP-DM to KDD is usually used enumerate the phases and discuss your techniques in achieving every tasks. In ML Process there are several frameworks which can be generally broken into several steps: such as Collection of Data, Data Wrangling, Model Building, Model Evaluation and Model Deployment. Please ensure that all methods, techniques, and approach will be based on scientific process and reputable references.

System Development Methodology

This section discusses the researchers' developmental methodology as the algorithm or data model integration in the software design. The models specify the various stages of the process and the order in which they are carried out. Software process model may also be reflected on this section.

Software Tools Used

This section discusses the programming language and software tools used in data modelling and system development. Elaborate the specific tasks and uses of each tool.

System Architecture

This section presents the system architecture of the developed system. Enumerate the functions and features highlight the integration of computing science solutions. Discuss the structure, behavior, and more views of a system.

Software Testing

This section discusses the software testing method and strategies to determine if the computing science problem was solved.